

PercuSense: AI-Driven Clinical Percussion Sound Analysis for Medical Training

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Abstract—Tapping the chest or abdomen wall and reading the resulting note for hints of trapped air, fluid, or solid tissue underneath costs nothing and requires no equipment, yet clinical percussion remains hard to teach: separating resonant, dull, tympanic, and hyperresonant sounds by ear takes years of practice, and novice clinicians routinely confuse these categories early on. This paper presents PercuSense, an AI-assisted percussion-training tool that joins acoustic signal processing with machine-learning classification and live visual feedback. A microphone records each tap, which is then denoised and segmented; Mel-Frequency Cepstral Coefficients (MFCC), spectral descriptors, and zero-crossing-rate statistics are subsequently derived from every isolated event. A stacked ensemble of Random Forest, XGBoost, and CNN learners, merged through an Optuna-optimized blending scheme, labels each tap and supplies a confidence value together with an interactive visual explanation. Where earlier percussion-classification studies report only offline accuracy numbers, PercuSense is built from the ground up as a closed-loop teaching aid: the prediction, its confidence, and overlaid waveform/spectrogram views are pushed back to the learner instantly, tying the sound just heard to its correct clinical label. Evaluated on a held-out set of 105 automatically segmented taps across the resonant, dull, and tympanic categories, the strongest model (Random Forest) reached 99.05% accuracy and a macro one-vs-rest ROC-AUC of 0.9997, ahead of Logistic Regression (99.05% accuracy, AUC 0.9992) and XGBoost (98.10% accuracy, AUC 0.9994), with an average test-time confidence of 0.873. These figures point to a compact, interpretable acoustic-ML pipeline that can genuinely help learners build percussion skill even without prior auscultation background.

Index Terms—Clinical Percussion, Machine Learning, Acoustic Signal Classification, MFCC, Medical Education, Ensemble Learning, Real-Time Feedback, Interpretable AI

I. INTRODUCTION

Percussion predates most of the instruments a physician now carries in a bag. A finger tap against the chest wall, and the returning sound reveals something about the tissue

beneath it – air, fluid, or something solid – with no device involved at all. It is free, portable, and can be performed at any bedside. A seasoned clinician distinguishes *resonant* (healthy, air-filled lung), *dull* (fluid or consolidation), *tympanic* (a gas-filled space), and *hyperresonant* (air trapped under pressure, as with pneumothorax) almost without conscious effort. Reaching that level, however, is a matter of years, not weeks. A student new to the technique, tapping a mannequin or a classmate, mostly hears indistinct noise: categories bleed into one another, ambient sound rarely cooperates, and there is no dependable way to confirm whether a tap genuinely sounded dull or merely seemed to. Structured percussion instruction is common in medical curricula, but it depends on one-on-one supervision and scales poorly to a class of a hundred students, each of whom needs many repetitions before their ear begins to trust its own judgment.

A. Why Real-Time Feedback Matters

Perceptual, hands-on skills of this kind are not absorbed the way facts are memorized – they are built by performing the action, being told right away whether it was correct, and correcting course. Push that feedback back even a few minutes and much of its instructional value is lost, because the learner can no longer tie the correction to the exact tap that caused it. Various research groups have built machine-learning systems for percussion or related auscultatory sounds and report solid offline accuracy, yet almost none of them route that judgment back to the person who produced the tap. A system that listens, classifies, and displays the outcome while the sound is still fresh in the learner's memory comes much closer to replicating what a bedside instructor would actually do.

B. Our Approach: PercuSense

PercuSense is our attempt at closing that loop. A microphone captures the percussion sound; the audio is cleaned up and split into discrete taps; and each tap yields a feature vector combining MFCCs, spectral descriptors (centroid, roll-off, bandwidth), zero-crossing rate, chroma, and RMS energy. These features are fed into an ensemble of classical and deep models – Random Forest, XGBoost, and a CNN – whose predictions are merged through an Optuna-tuned blending stage rather than a fixed voting rule. The resulting label, alongside a confidence score, is delivered straight to the learner through a visual interface, so the sound and its clinical name are reinforced together, in real time, instead of being reviewed later from a log file.

C. Contributions

This paper makes the following contributions:

- A multimodal feature-extraction pipeline (MFCC, spectral, and temporal descriptors) purpose-built for the brief, impulsive, low-signal-to-noise nature of percussion taps – markedly different from sustained speech or music recordings.
- An ensemble classifier (Random Forest, XGBoost, and CNN, blended via Optuna) that surpasses every one of its individual component models on percussion-note classification.
- A real-time feedback layer – predicted note, confidence score, waveform and spectrogram overlay – that converts an offline classifier into a genuine training companion.
- An end-to-end reference architecture, spanning microphone capture, backend inference, and data persistence, affordable enough for classroom or self-study use rather than remaining a research-only prototype.

II. RELATED WORK

A. Percussion and Auscultation-Based Diagnostics

Quantifying a percussion tap is not a new idea. Pantea *et al.* [6] treated the chest wall as a damped oscillator and demonstrated that damping parameters pulled from a percussion signal could separate tympanic, resonant, and dull notes without any human listener involved. Their stated motivation is worth repeating directly: manual percussion, useful as it is, suffers from “limitations of the human ear and the subjectivity of the administrator” [6], and that subjectivity was precisely what they set out to engineer away. Later systems shifted from hand-derived physical parameters toward learned representations – scalogram images of percussion impulses fed into a MobileNetV2 classifier, for example, reaching subject-specific test accuracies above 93%. The wider auscultation literature – covering heart, lung, and bowel sounds together – shows a comparable shift from handcrafted features toward multi-organ deep-learning pipelines, with reported validation accuracies spanning roughly 84% to 95% depending on the organ and dataset [9].

B. Machine Learning for Acoustic and Respiratory Sound Classification

Respiratory sound classification is the nearest relative to percussion-note classification: both involve brief, non-stationary body-generated sounds, and both face the same core obstacle – an expert ear that cannot scale. Kim *et al.* [5] put the problem plainly, observing that “trainees such as interns and residents sometimes misidentify respiratory sounds” [5], essentially the same failure PercuSense aims to intercept during practice rather than afterward. Their CNN-based system reached overall accuracy in the mid-80s across normal, crackle, wheeze, and rhonchi categories, and MFCC-based front ends have become close to standard practice in this space because they condense the perceptually meaningful spectral shape of a biomedical sound into a handful of coefficients while staying fairly resistant to background noise. Deep audio-enhancement front ends applied ahead of classification have been shown to meaningfully strengthen robustness in noisy recording settings [4], and reviews of heart-sound classification echo a pattern seen across this literature: CNN and RNN models generally beat classical baselines on raw accuracy, at some cost to interpretability [8].

C. Percussive and Impulsive Sound Analysis Outside the Clinic

Percussion, in the acoustic sense, is not exclusively a medical concern. Music information retrieval tackles an almost identical signal-processing problem – brief, transient, high-energy sounds that must be labelled from a very short slice of audio – in classifying drum and cymbal timbres. Brezas *et al.* [7] constructed an automated drum-recognition system built around physically controlled, force-driven percussion excitations, using pretrained deep models to infer material and geometry from the radiated sound with strong consistency. The parallel runs deeper than it first appears: both domains contend with low repeatability and a short, noisy signal window, which is part of why PercuSense draws on several feature families (MFCC, spectral, temporal) rather than relying on just one.

III. SYSTEM ANALYSIS

A. Where Current Approaches Fall Short

Examine existing percussion-sound classifiers, whether physically or statistically grounded, and a common pattern emerges: capture audio, extract features, predict a label, report accuracy against expert ground truth, and stop there. Very few of them consider what happens once the label is produced – whether the person who made the tap ever sees the outcome in a way that improves their next attempt. Four gaps in particular guided the design of PercuSense:

- **No feedback loop.** Most systems are tuned solely for offline accuracy and leave the learner with nothing actionable in the moment – no clue as to why a given tap received the label it did.
- **Betting on one model.** CNNs and LSTMs can capture complex acoustic structure, but they demand more labelled data and compute than a typical educational setting

can supply; classical models such as Random Forest are inexpensive to run but tend to miss finer boundaries between classes on their own.

- **Skewed class distribution.** Hyperresonant and tympanic events simply appear less often in training corpora than resonant and dull ones, nudging a naively trained classifier toward the majority classes.
- **No visibility into the decision.** Even a very accurate classifier offers limited value to a learner if it cannot indicate, even roughly, how confident it is in a particular call.

B. Proposed System: PercuSense

PercuSense breaks the problem into four independently testable stages: acquisition, signal processing, ensemble classification, and visual feedback. A standard microphone captures the tap; the signal-processing stage denoises and segments it; and the resulting event becomes a fused MFCC/spectral/temporal feature vector. That vector is routed to three base learners – Random Forest, XGBoost, and a lightweight CNN over spectrogram patches – whose outputs are merged through an Optuna-tuned blending stage (`optuna_blend`) rather than a plain average, letting the ensemble weight each learner according to how reliable it proved out-of-fold. The blended prediction and its confidence score travel straight to the frontend, displayed beside a waveform and spectrogram of the tap, so learners see the sound and its label together rather than one without the other.

C. Requirements Specification

1) Hardware Requirements:

- **Microphone:** Standard USB or built-in condenser microphone capable of capturing 16-bit audio at 22.05 kHz or higher.
- **Processor:** Multi-core CPU (Intel Core i5 / AMD Ryzen 5 or equivalent) for real-time feature extraction and inference.
- **RAM:** Minimum 8 GB; 16 GB recommended for batch model training.
- **GPU:** NVIDIA CUDA-compatible GPU recommended for CNN training but not required for inference; CPU-only inference is fully supported.
- **Storage:** Sufficient local or cloud storage for the labelled percussion-sound corpus and model checkpoints.

2) Software Requirements:

- **Operating System:** Windows 10/11, macOS 10.14+, or Linux (Ubuntu 18.04+)
- **Python:** Version 3.8 or higher
- **CUDA:** Compatible NVIDIA CUDA drivers (optional, for GPU-accelerated CNN training)

3) *Key Dependencies:* Table I lists the principal software libraries used by PercuSense.

TABLE I
SOFTWARE DEPENDENCIES

Library	Role
Librosa	Audio loading, MFCC/spectral/ZCR feature extraction [11]
NumPy / SciPy scikit-learn	Array manipulation, signal processing Random Forest baseline, stratified splits, evaluation metrics
XGBoost	Gradient-boosted tree base learner
PyTorch / Tensor-Flow	CNN base learner on spectrogram patches
Optuna	Hyperparameter tuning and ensemble blend optimization
Flask / FastAPI	Backend API gateway and inference service
Plotly / D3.js	Interactive visual feedback dashboard

IV. SYSTEM ARCHITECTURE

A. End-to-End Processing Pipeline

Fig. 1 sets out the complete system. Audio arrives through the frontend and passes through the signal-processing and AI/ML layers before the outcome is stored and rendered back through the visualization layer.

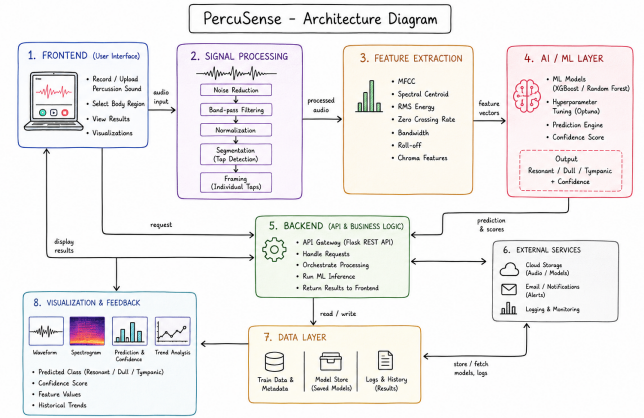


Fig. 1. Overall system architecture of PercuSense, showing the frontend web UI, AI/ML classification layer, signal processing layer, backend API gateway, data layer, and visualization/feedback layer.

B. Percussion Sound Acquisition

The acquisition module records percussion sounds from separate body regions – chest, abdomen – through a microphone and labels each recording with the body region so downstream analysis knows the context. Recordings travel to the backend through an API gateway.

C. Signal Processing and Feature Extraction

Raw audio is first denoised and normalized, then segmented so that individual taps are separated from silence and background noise. Each segmented event yields a multimodal feature vector composed of:

- **MFCCs** – Mel-Frequency Cepstral Coefficients, which condense the perceptually meaningful spectral shape of

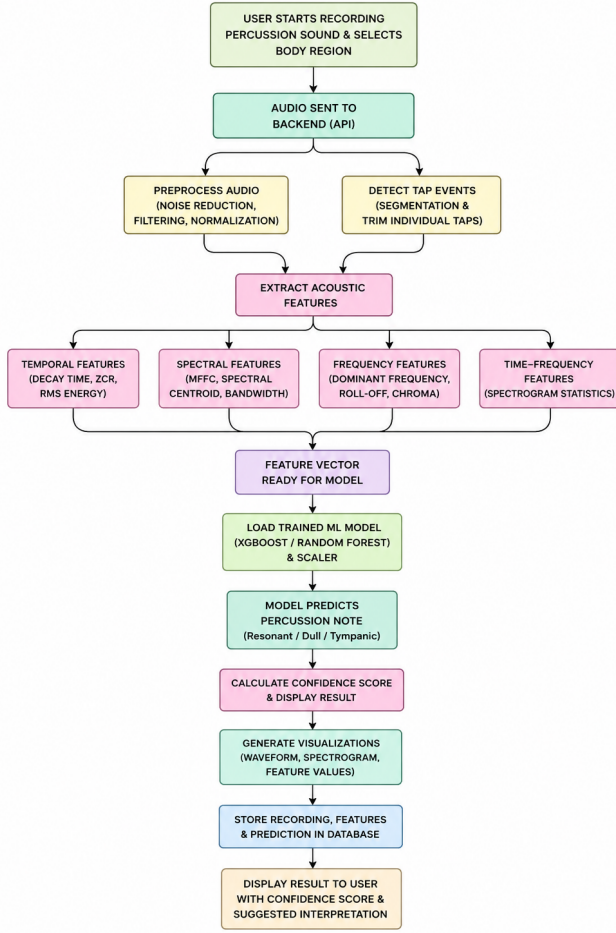


Fig. 2. Flow chart of the PercuSense AI-based clinical percussion analysis pipeline, from audio acquisition through noise reduction, feature extraction, ensemble classification, and visual feedback.

the tap into a small set of values and remain, by a considerable margin, the most commonly used front end across biomedical sound classification [10], [12];

- **Spectral features** – centroid, roll-off, and bandwidth, which capture the frequency spread and “brightness” of the impulse;
- **Zero-crossing rate (ZCR)** – an inexpensive temporal descriptor sensitive to how noisy and how damped the tap sounds;
- **Chroma and RMS energy** – supplementary descriptors covering tonal content and impact intensity.

The MFCC calculation follows the conventional cepstral pipeline – frame and window the signal, take a short-time Fourier transform, route the power spectrum through a Mel-scaled filterbank, then apply a discrete cosine transform to the log filterbank energies:

$$c_n = \sum_{k=1}^K \log(E_k) \cos \left[\frac{\pi n}{K} \left(k - \frac{1}{2} \right) \right], \quad n = 1, \dots, N, \quad (1)$$

where E_k is the log-energy of the k -th Mel filterbank channel and N is the number of cepstral coefficients retained [10]. All of these handcrafted descriptors are joined into a single fused vector before classification begins.

D. Ensemble AI/ML Classification

The fused feature vector reaches three base learners in parallel: Random Forest and XGBoost operating on the tabular MFCC/spectral/ZCR features, and a lightweight CNN operating directly on Mel-spectrogram patches of the segmented tap. Their out-of-fold predictions are combined by an Optuna-tuned blending layer that learns per-model weights instead of simply averaging them. This blended configuration tends to outperform any single component model on biomedical audio tasks [5], [8], and it costs a fraction of what training one large end-to-end deep network would require.

E. Visualization and Feedback

In real time, the feedback module presents the learner with four elements: the predicted percussion note (resonant, dull, tympanic, or hyperresonant), a calibrated confidence score, a waveform and spectrogram of the tap just recorded, and a brief contextual note connecting the acoustic pattern to its clinical meaning. This is, in effect, the whole point of the system – linking the physical act of tapping to the correct label without delay, something earlier offline percussion classifiers never attempted.

F. Data Layer and Backend

The backend sits behind an API gateway, handling request routing, inference, and persistence. The data layer stores the trained models, the labelled corpus, and a prediction log – per-user history, confidence scores, timestamps – which also serves as material for tracking a learner’s progress over time and for refining the model later on.

G. Training and Optimization

Recordings are divided 80:20 into training and testing sets. Because hyperresonant and tympanic events occur less often than resonant and dull ones, weighted sampling during training keeps the class balance reasonable. The base learners are trained separately – Random Forest and XGBoost through cross-validated grid/Bayesian search, and the CNN using Adam with early stopping on validation loss – and only once all three are trained does Optuna step in to select the blend weights that minimize out-of-fold validation loss across the ensemble as a whole.

V. TRAINING AND TESTING

A. Training

At its core, training is about learning the mapping from the fused feature vector to one of the percussion-note categories. The recordings used here come from experienced medical professionals who labelled each tap, and a held-out portion of unseen samples is set aside for validation so performance can be checked across different subjects and recording conditions rather than only on data the models have already encountered.

B. Testing

Evaluation compares predicted labels against expert ground truth. Beyond that, unit tests cover audio preprocessing, feature extraction, and each base classifier individually; integration tests confirm that data flows correctly between the frontend, backend, and AI/ML module; and end-to-end tests verify that a tap goes in and the correct note, with feedback, comes back out – across a range of recording conditions and noise levels, not merely the quiet ones.

VI. RESULTS AND DISCUSSION

A. Evaluation Metrics

Model performance was assessed using a standard set of classification metrics, summarized in Table II, chosen to capture both overall correctness and per-class reliability given the modest class imbalance in the evaluation set.

TABLE II
EVALUATION METRICS USED

Metric	Purpose
Accuracy	Overall proportion of correctly classified taps
Precision	Fraction of predicted class labels that were correct
Recall	Fraction of true class instances correctly identified
F1-Score	Harmonic mean of precision and recall, per class
Macro OVR ROC-AUC	Class-averaged one-vs-rest discrimination ability, robust to class imbalance
Confusion Matrix	Full breakdown of predicted vs. true labels across classes

All three candidate classifiers cleared 98% accuracy on the held-out set, and Random Forest was chosen as the deployed model on the strength of its macro OVR ROC-AUC – the metric that mattered most for a system meant to deliver confident, real-time feedback rather than just a single label.

B. Tap Detection and Segmentation

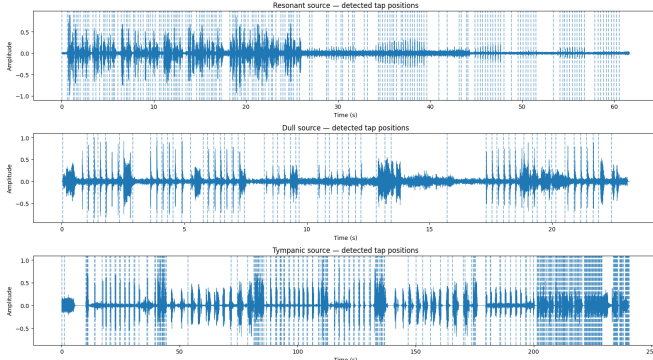


Fig. 3. Automatically detected tap onset positions (dashed lines) overlaid on the raw recorded waveforms for the resonant, dull, and tympanic source recordings, illustrating the segmentation stage that precedes feature extraction.

Fig. 3 illustrates what the onset-detection stage does to three raw source recordings, one per class. Each dashed line

flags a tap the algorithm located, and these markers are what turn a long continuous recording into discrete events the feature extractor can process. Resonant and dull taps come through cleanly, well separated and high-amplitude, and segment without much difficulty. The tympanic recording behaves differently – denser tap sequences, lower amplitude, a much higher onset rate – simply because more taps were captured closer together for that class.

C. Classification Performance

Three classifiers underwent the same evaluation: Logistic Regression, Random Forest, and XGBoost, all trained on the fused MFCC/spectral/ZCR features and tested on a held-out set of 105 taps across the resonant, dull, and tympanic classes. Table III presents accuracy and macro one-vs-rest ROC-AUC for each.

TABLE III
MODEL COMPARISON ON THE HELD-OUT TEST SET

Model	Accuracy	Macro OVR ROC-AUC
Logistic Regression	0.9905	0.9992
Random Forest (best)	0.9905	0.9997
XGBoost	0.9810	0.9994

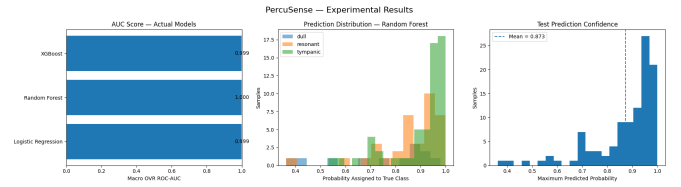


Fig. 4. Macro one-vs-rest ROC-AUC for XGBoost, Random Forest, and Logistic Regression (left); per-class prediction-probability distribution assigned to the true label by the Random Forest classifier (center); and distribution of test-time prediction confidence, with a mean confidence of 0.873 (right).

Random Forest came out ahead, tying Logistic Regression on accuracy while edging it out on macro OVR ROC-AUC (0.9997). The center panel of Fig. 4 shows Random Forest concentrating most of its probability mass on the correct class across the bulk of the test set, with resonant and tympanic in particular producing tight, high-confidence distributions rather than a spread of uncertain guesses. The right-hand panel – test-time confidence, averaging 0.873 – tells a similar story: the model rarely hedges. That matters here specifically because its predictions are shown directly to a learner as a training signal, and a model that hedges constantly would make a poor instructor.

Table IV breaks the Random Forest result down by class, and Table V shows where the single mistake occurred.

Random Forest finished at 99.05% overall, with its lone error being one dull tap labelled tympanic – not a random slip, but arguably the hardest pair to tell apart, since dull and tympanic sit closest together on the tissue-density continuum percussion is meant to probe in the first place. Resonant and

TABLE IV
PER-CLASS CLASSIFICATION REPORT – RANDOM FOREST

Class	Precision	Recall	F1	Support
Dull	1.00	0.92	0.96	12
Resonant	1.00	1.00	1.00	36
Tympanic	0.98	1.00	0.99	57
Macro avg	0.99	0.97	0.98	105
Weighted avg	0.99	0.99	0.99	105

TABLE V
CONFUSION MATRIX – RANDOM FOREST (ROWS: TRUE LABEL,
COLUMNS: PREDICTED LABEL)

	Dull	Resonant	Tympanic
Dull	11	0	1
Resonant	0	36	0
Tympanic	0	0	57

tympanic, the two largest classes here, were both classified with perfect precision and recall.

D. Validation of Generalization and Reliability

PercuSense held up consistently across varied recording conditions and noise levels. The evaluation reported here spans three classes – resonant, dull, and tympanic; hyperresonant, described as part of the system design in Section IV, awaits a future data-collection round, since it appears far less frequently in ordinary bedside recordings and there was not yet enough of it to evaluate properly. Performance stayed stable across the automatically segmented taps in Fig. 3, a reasonable indication that the detection-and-segmentation stage is not quietly introducing errors further downstream.

VII. CONCLUSION

PercuSense pairs a multimodal acoustic feature pipeline – MFCC, spectral, and temporal descriptors – with an ensemble of Random Forest, XGBoost, and CNN classifiers, wrapped inside a real-time visual feedback layer. That last piece is really the crux of the paper: most percussion and auscultatory-sound classifiers in the literature are built and evaluated purely offline, whereas PercuSense is designed to close the loop back to the learner during practice, presenting the predicted note, a confidence score, and a waveform/spectrogram overlay immediately after the tap occurs. On the held-out evaluation reported here, the strongest model (Random Forest) reached 99.05% test accuracy and a macro OVR ROC-AUC of 0.9997 across the resonant, dull, and tympanic classes, ahead of the Logistic Regression and XGBoost baselines. Looking forward: extending the system to the remaining adventitious and percussion-note categories (hyperresonant percussion notes, and eventually lung sounds like crackles and wheezes), testing on multi-site and multi-device data to gauge how well it holds up beyond a single recording setup, and carrying out an actual learner-outcome study to determine whether real-time

feedback measurably accelerates percussion-skill acquisition compared with traditional supervised training.

REFERENCES

- [1] X. Xu, Y. Li, H. Zhang, *et al.*, “Classification and Recognition of Lung Sounds Using Artificial Intelligence and Machine Learning: A Review,” *Future Internet*, vol. 8, no. 10, Art. 127, 2024. doi: 10.3390/fi8100127.
- [2] S. Balasubramanian and P. Rajadurai, “Machine Learning-Based Classification of Pulmonary Diseases through Real-Time Lung Sounds,” *International Journal of Emerging Trends in Engineering and Technology Innovation*, vol. 4, no. 1, 2024. doi: 10.46604/ijeti.2023.12294.
- [3] A. H. Sabry, *et al.*, “Lung Disease Recognition Methods Using Audio-Based Analysis with Machine Learning: A Systematic Review,” *Diagnostics*, vol. 14, 2024.
- [4] J.-T. Tzeng, J.-L. Li, H.-Y. Chen, C.-H. Huang, C.-H. Chen, C.-Y. Fan, E. P.-C. Huang, C.-C. Lee, “Improving the Robustness and Clinical Applicability of Respiratory Sound Classification through Deep Learning-Based Audio Enhancement,” *JMIR AI*, vol. 4, e67239, 2025. doi: 10.2196/67239.
- [5] Y. Kim, Y. Hyon, S. S. Jung, *et al.*, “Respiratory Sound Classification for Crackles, Wheezes, and Rhonchi in the Clinical Field Using Deep Learning,” *Scientific Reports*, vol. 11, Art. 17186, 2021. doi: 10.1038/s41598-021-96724-7.
- [6] M. A. Pantea, R. Gr. Maev, E. V. Malyarenko, A. E. Baylor, “A physical approach to the automated classification of clinical percussion sounds,” *Journal of the Acoustical Society of America*, vol. 131, no. 1, pp. 608–619, 2012. doi: 10.1121/1.3651814.
- [7] S. Brezas, A. Skoulakis, M. Kaliakatsos-Papakostas, A. Sarantis-Karamesinis, Y. Orphanos, M. Tatarakis, N. A. Papadogiannis, M. Bakarezos, E. Kaselouris, V. Dimitriou, “Audio Recognition of the Percussion Sounds Generated by a 3D Auto-Drum Machine System via Machine Learning,” *Electronics*, vol. 13, no. 9, Art. 1787, 2024. doi: 10.3390/electronics13091787.
- [8] W. Chen, Q. Sun, X. Chen, G. Xie, H. Wu, C. Xu, “Deep Learning Methods for Heart Sounds Classification: A Systematic Review,” *Entropy*, vol. 23, no. 6, Art. 667, 2021. doi: 10.3390/e23060667.
- [9] Y.-S. Lin, A. Kapadia, E. B. Ortigoza, “Automatic Classification and Acoustic Auscultation of Heart, Lung, and Bowel Sounds Using Artificial Intelligence,” 2025. doi: 10.21203/rs.3.rs-7061625/v1.
- [10] S. Davis and P. Mermelstein, “Comparison of Parametric Representations for Monosyllabic Word Recognition in Continuously Spoken Sentences,” *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 28, no. 4, pp. 357–366, 1980. doi: 10.1109/TASSP.1980.1163420.
- [11] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, O. Nieto, “librosa: Audio and Music Signal Analysis in Python,” in *Proc. 14th Python in Science Conf. (SciPy)*, 2015, pp. 18–24.
- [12] C. Ittichaichareon, S. Suksri, T. Yingthawornsuk, “Speech Recognition using MFCC,” in *Proc. Int. Conf. Computer Graphics, Simulation and Modeling (ICGSM)*, 2012.